

Percolation Weight and Thermodynamic Stability Across a 186-Compound Dataset

Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis

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Update, August 15, 2026: missions #3–#5 added (dimensionality/min-cut, antibonding population, reaction ICOHP), see §11. The body of this report below (through the conclusion and appendices) documents the original campaign 2 (186 compounds, percolation weight only) unmodified.

Update, August 16, 2026: dataset extended to 349 compounds (89-compound `extension4` batch, alkali/alkaline-earth binaries), missions #3–#5 reprocessed at the new scale, and the periodic min-cut correlation’s collapse at that scale diagnosed — see §12.

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1 Objective and Problem Statement

The first report (6 toy compounds) established that the percolation descriptor computed by `percolation_path.py` diverges sharply from classic ICOHP aggregates (sum, mean, min, max), but without a sample large enough to statistically test whether that difference is *useful*: does the percolation descriptor carry predictive information about thermodynamic stability (`energy_above_hull`) that the classic aggregates don't already carry?

This second campaign builds a 186-compound dataset (180 new + the 6 pilot compounds) split into three families:

- **exp_stable**: experimental (ICSD-referenced) compounds on the Materials Project thermodynamic convex hull;
- **exp_metastable**: experimental metastable compounds ($0 < E_{\text{hull}} \leq 100$ meV/atom);
- **theo_metastable**: theoretical (never experimentally observed) metastable compounds ($0 < \bar{E}_{\text{hull}} \leq 200$ meV/atom).

The `exp_metastable` / `theo_metastable` split serves as a proxy for kinetic persistence: an experimentally observed metastable compound has, by definition, survived long enough to be synthesized and characterized, unlike a theoretical compound of comparable thermodynamic metastability.

2 Methodology

2.1 Dataset Selection

Materials Project query (`mp_dataset/select_campaign.py`): unary and binary chemical systems, non-magnetic ($|\text{total magnetization}| \leq 0.01 \mu_B/\text{formula unit}$ — MP's categorical `magnetic_ordering` field is "Unknown" for $\sim 98\%$ of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements, ≤ 20 atoms/cell, 60 compounds per family with a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system.

2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by `fetch_candidates.py`'s `classify()` function (**reused unchanged**, per the campaign's instructions), from `is_metal` and elemental composition. This is a secondary analysis axis here, not the primary sampling axis (unlike the campaign's originally proposed plan, replaced at the user's request with the stable/metastable scheme above).

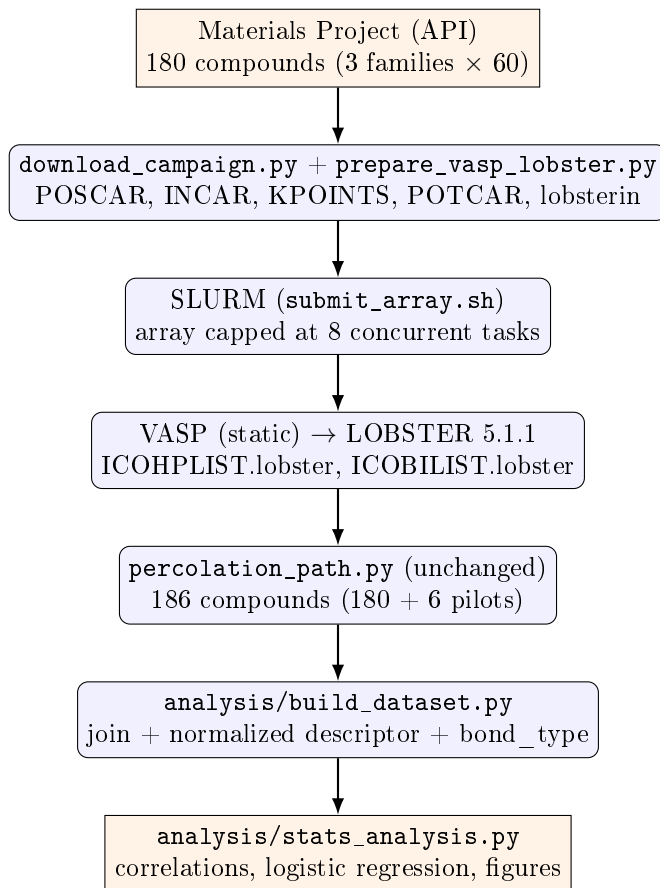
2.3 VASP + LOBSTER Computation

Same template as the pilot campaign (static run, `ISYM=-1`, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPRelaxSet`), with two improvements: `NBANDS` is no longer a fixed value but derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

2.4 Statistics

Spearman correlation (no reason to expect a linear relationship) between percolation weight (raw and normalized by $|\text{ICOHP}_{\min}|$) and `energy_above_hull`, overall and per bond type; same for the classic ICOHP aggregates, for direct comparison; a reference logistic regression (stable vs. metastable, 2–3 features, 5-fold cross-validation) for a reference AUC. **No SISSO, no symbolic regression**: premature with a single candidate descriptor at this sample size (see §9).

3 Workflow Diagram



4 Environment

Same as the pilot campaign (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib` (see `requirements.txt`).

5 Dataset

Family	n	E_{hull} range (eV/atom)
exp_stable	63	0.0000 (by construction)
exp_metastable	63	0.0005 – 0.0907
theo_metastable	60	0.0013 – 0.1991

Bond type (<code>classify()</code>)	<i>n</i>
metallic	88
covalent	23
ionic	9
unclassified	66

35% of compounds (66/186) fall outside `classify()`'s three categories — mostly transition-metal pnictides/chalcogenides and hydrogen-containing compounds, a real limitation of the reused heuristic, documented but not fixed here.

6 Results

6.1 Correlations (Spearman)

Group	Metric	<i>n</i>	ρ	<i>p</i>
all	percolation weight (raw)	186	0.064	0.383
all	percolation weight (normalized)	186	0.071	0.339
metallic	percolation weight (raw)	88	0.191	0.075
metallic	percolation weight (normalized)	88	0.222	0.038
ionic*	percolation weight (raw)	9	−0.402	0.284
covalent	percolation weight (raw)	23	0.165	0.453
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

* *n* < 15: do not over-interpret.

In the one subgroup where any result reaches nominal significance (metallic, *n* = 88), the normalized percolation weight (ρ = 0.222, *p* = 0.038) and the ICOHP sum (ρ = 0.223, *p* = 0.037) are statistically indistinguishable in strength. **The percolation descriptor does not demonstrate additional predictive power over the classic aggregates on this dataset** — a more precise answer than the previous (6-compound) report's "qualitatively different" framing, and a more directly actionable one.

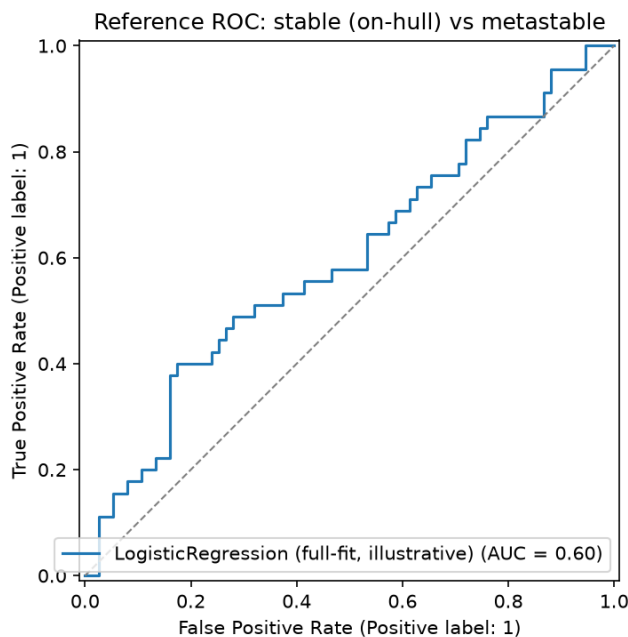
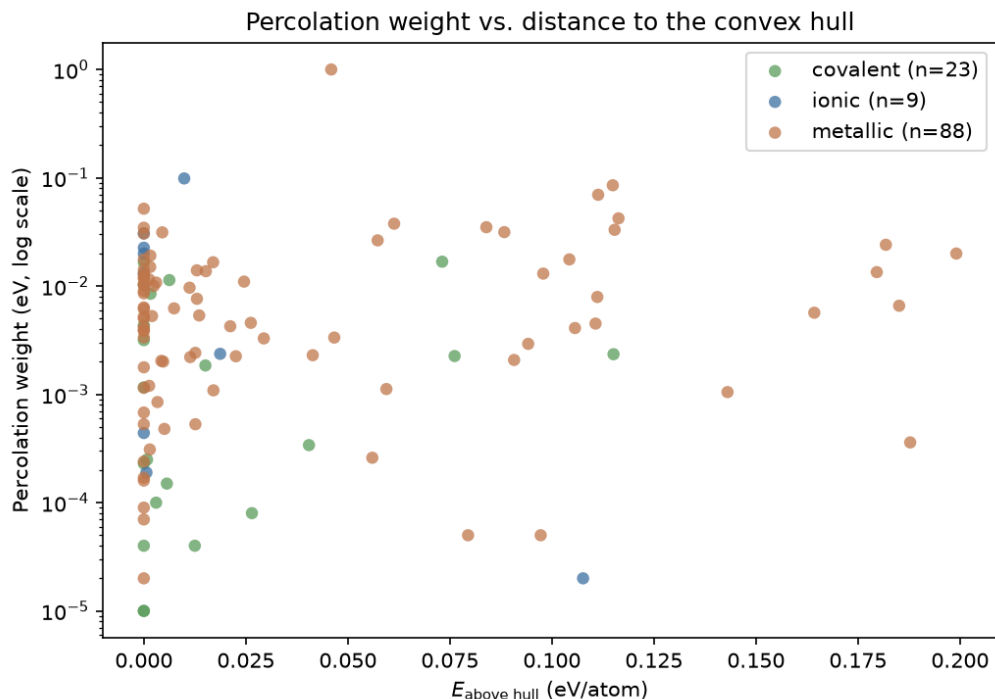
6.2 Reference Logistic Regression

Stable (on-hull) vs. metastable, features: normalized percolation weight, mean ICOHP, bond-type dummies (*n* = 120 after dropping unclassified compounds).

AUC (5-fold cross-validation)	0.496 ± 0.141
per-fold AUC	0.556, 0.430, 0.252, 0.637, 0.607

AUC $\approx$ 0.5: chance level. The wide spread across folds (0.25–0.64) is itself informative: it indicates the absence of a stable, learnable signal at this sample size, not merely a "weak but real" one.

6.3 Figures



7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the 178 SLURM-array-submitted jobs failed on the first pass, all diagnosed and fixed at the source:

- **Al12W, BW, WO2, TeW, TiW** (all tungsten-containing): the local mirror's `W_sv` POTCAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).

- **A112W** failed again after that fix, this time with a VASP SIGSEGV — exactly the known issue flagged in the mission brief (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`'s per-compound SLURM template. Fixed; recovered on retry.

Final success rate: **186/186**.

8 Compute Cost

	VASP	LOBSTER
Mean (s)	124	397
Median (s)	72	145
Max (s)	1440	6675
Sequential total	6.4 h	19.1 h

Actual elapsed wall-clock time (8-way concurrency): ~ 3.5 h, within the 6 h overnight budget.

9 Limits and Next Steps

- **Sample size:** 186 overall, but 88/23/9 once split by bond type — the "ionic" correlations ($n = 9$) are not interpretable.
- **Primitive vs. conventional cell** (carried over from the pilot report): not addressed here. Most likely explanation for the weak observed correlations — the strongest bond can sit entirely inside the primitive cell and contribute to no non-contractile cycle at all, biasing the measured "weakest link."
- **No true dynamical "unstable" class:** `theo_metastable` is a proxy (never synthesized), not a phonon-calculation result.
- **classify() coverage:** 35% of compounds unclassified.
- **Why not SISSO yet:** a single candidate descriptor (the percolation weight, in two algebraically related variants), compared against aggregates that are themselves functions of the same underlying ICOHP values — no meaningful combinatorial search space yet. Justifying SISSO would require: (a) several conceptually novel candidate descriptors, (b) per-stratum $n \geq 50$ –100, (c) a clearer signal than $\text{AUC} \approx 0.5$.

10 Conclusion

The 186-compound dataset was built, computed (VASP+LOBSTER), and analyzed end to end, with two compute failures diagnosed and fixed at the source. The headline result is a *negative* but quantified and honest one: at this sample size, the percolation descriptor shows no significant correlation with thermodynamic stability, and demonstrates no advantage over classic ICOHP aggregates. The priority actions (§9) — especially conventional-cell reprocessing and a targeted extension of the metallic subsample — are likely to move this result before a SISSO-class tool is warranted.

Source code: <https://github.com/John-Akapulco/viability>

Data: `analysis/percolation_vs_hull.csv`

Detailed report: `analysis/REPORT.md`

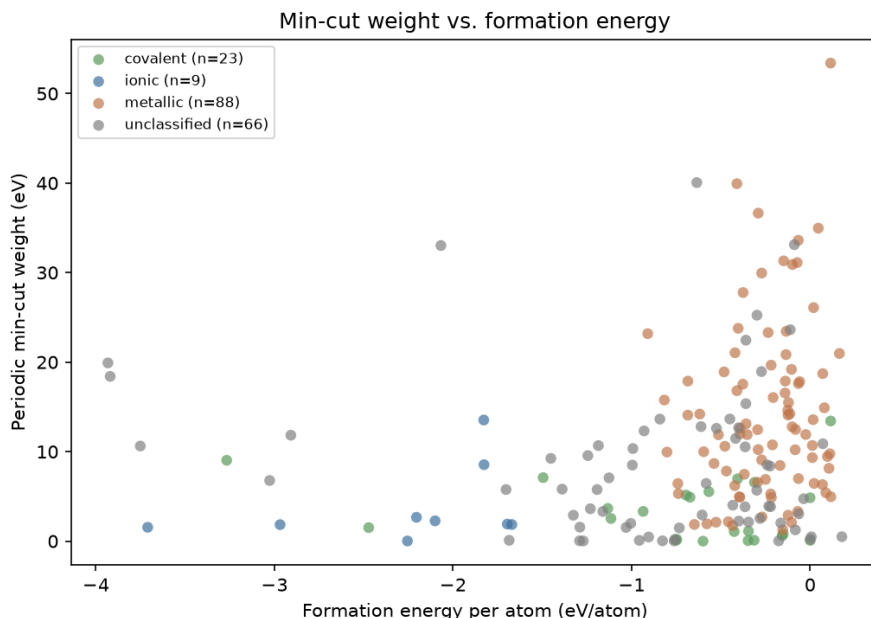
11 Campaign 2 follow-ups (missions #3 to #5)

Three further descriptors were built in later sessions, all tested against `formation_energy_per_atom` (introduced in mission #3 as an alternative target to `energy_above_hull`, generally more correlatable) rather than the hull. Missions #3 and #4 reuse the 186-compound dataset above unmodified; mission #5 draws in part on a **74-compound extension campaign** (`extension_*`: elemental references, ionic compounds and polymorphs outside the main set), not reprocessed into this report’s appendices — see the repository’s `README.md` for its full detail. Numbers in this subsection are as of August 15, 2026, at the 186/192-compound scale; §12 reprocesses all three at the 349-compound scale reached the following day and should be read alongside this subsection, not instead of it.

A pattern repeats across all three missions: each new descriptor reaches a statistically significant *global* correlation, stronger than anything the percolation weight ever achieved (§9) — but in all three cases that global correlation turns out to be substantially, or entirely, a clustering effect between bond types or between metals and gapped compounds, rather than a robust relationship *within* a homogeneous group.

11.1 Mission #3: network dimensionality and periodic min-cut

Two new descriptors, `percolation_path.py` left unchanged: `network_dimensionality.py` (0D–3D classification via a relative bond-strength threshold + breadth-first search) and `periodic_mincut.py` (minimum cut via `networkx` max-flow, separability rather than traversability). Headline: the normalized min-cut is the **first descriptor in the project to reach a significant global correlation** ($\rho = 0.285$, $p = 1.0 \times 10^{-4}$, $n = 186$), likely driven in part by between-bond-type clustering; dimensionality alone does not separate `formation_energy_per_atom` (Kruskal–Wallis $p = 0.19$).



11.2 Mission #4: antibonding population near the Fermi level

A distinct question: not the integrated ICOHP (one number per bond), but *how* COHP is distributed in energy — the occupied antibonding fraction just below the Fermi level/VBM, by analogy with Peierls/Jahn–Teller electronic instabilities (new module `cohp_extraction.py`, validated by exact cross-check against `ICOHPLIST.lobster` on the 6 pilot compounds). Headline:

$\rho = -0.328$, $p = 5.0 \times 10^{-6}$ ($n = 186$) — the strongest global correlation in the project at the time. The sign is *opposite* the naive Peierls/Jahn–Teller reading. Diagnostics: the effect is substantially driven by metal/gapped clustering, but the **bond_type=covalent** subgroup ($n = 23$) survives stratification ($p = 0.018$ – 0.029) — the only subgroup in the project, to date, to hold up under such a test.

11.3 Mission #5: reaction ICOHP

An ICOHP analog of **formation_energy_per_atom**: is a compound’s total ICOHP “cheaper” than the same atoms in elemental reference form (new module **reaction_icohp.py**, reaction balanced via **pymatgen.analysis.reaction_calculator**)? Extended to 192 compounds using the 74-compound extension campaign (62 elemental references computed). Headline: $\rho = 0.370$, $p = 3.9 \times 10^{-7}$ ($n = 177$) — numerically the strongest global correlation in the project to date, this time with an intuitive sign (less bonding than the elemental state $\Rightarrow$ less stable formation). But **no bond_type subgroup survives stratification** (Kruskal–Wallis $p = 2.1 \times 10^{-6}$ for the descriptor, $p = 5.0 \times 10^{-8}$ for the target, across the three strata) — unlike mission #4, the clustering effect here explains the entire observed signal. Against **energy_above_hull** the correlation is absent ($\rho = 0.09$, $p = 0.20$).

11.4 Comparative summary (186/192-compound scale)

All four descriptors tested against **formation_energy_per_atom**, on the same common subsample ($n = 177$, the intersection of campaign 2’s 186 compounds and mission #5’s 192 computable rows):

Descriptor	n	ρ	p
Reaction ICOHP (mission #5)	177	0.370	3.9×10^{-7}
Antibonding population, normalized (mission #4)	177	−0.337	4.5×10^{-6}
Periodic min-cut, normalized (mission #3)	177	0.313	2.3×10^{-5}
Percolation weight, normalized (mission #1)	177	0.104	0.170

Ranking by raw correlation strength (ρ) is not the same as ranking by reliability: mission #5 leads numerically but is also the only one where *no* subgroup survives stratification (above), whereas mission #4 keeps a significant covalent subgroup.

12 August 16 update: the extension4 campaign, all three missions reprocessed, and the min-cut collapse

12.1 The extension4 campaign

89 new compounds (**mp_dataset/download_extension4.py**): binary compounds of an alkali (Li/Na/K/Rb/Cs) or alkaline-earth (Be/Mg/Ca/Sr/Ba) metal against N/O/F/P/S/Cl, sourced from Materials Project. Up to 2 compounds picked per (cation, anion) chemical system across 44 systems: an experimentally-known polymorph (preferring formulas already elsewhere in the dataset, to build genuine same-composition polymorph groups — the dataset had only 8 such groups before this batch) and the highest-**energy_above_hull** theoretical entry available for that formula, a deliberate far-from-equilibrium stress test. Selection was computed once and frozen to **extension4_candidates.json** for reproducibility (scale made hand-curation, the convention for every earlier extension batch, impractical). All 89 VASP+LOBSTER jobs completed unattended (a self-healing watcher restarted any job that left the SLURM queue without producing output; zero restarts were actually needed). Dataset: 260 $\rightarrow$ 349 compounds.

Separately, the schema-driven `reaction_analysis/` package (superseding the ad hoc `reaction_icohp.py` module long-term) was extended with an **endobondic/exobondic bonding-kinetics classifier** following Reitz & Dronskowski (ic-2026-04181q): a reaction’s ΔICOHP (products – reactants) being positive (“endobondic”) means breaking the reactant’s bonds costs more than the products’ bonds recover — a kinetic barrier that can keep an exothermically-favored decomposition from actually happening, explaining why some very exothermic compounds are nonetheless observed to exist. Validated against all 7 worked numerical examples in the manuscript; not yet connected to this project’s own LOBSTER data for classification, an explicit next step, not attempted this session.

Every reaction’s per-bond-type ICOHP values were hand-transcribed from the manuscript text (`tests/fixtures/reitz_dronskowski_cases.yaml`), independent of this project’s own CSP data, then run end to end through `balance.py/delta.py/classify.py`. All 7 match the manuscript’s reported ΔICOHP within a few kJ/mol (attributable to the manuscript’s own rounding of its reported per-bond eV values) and every `BondingLabel` matches:

Reaction	Manuscript (kJ/mol)	Computed (kJ/mol)	Label Ms.	Computed
$\text{Pb}(\text{N}_3)_2 \rightarrow \text{Pb} + 3\text{N}_2$	+1345	+1344.3	endo	endo
$\text{S}_4\text{N}_2 \rightarrow \frac{1}{2}\text{S}_8 + \text{N}_2$	+258	+258.5	endo	endo
$\text{S}_4\text{N}_4 \rightarrow \frac{1}{2}\text{S}_8 + 2\text{N}_2$	+399	+397.3	endo	endo
$\text{ZnSn} \rightarrow \text{Zn} + \text{Sn}$	−337	−336.6	exo	exo
$\text{CaO}[\text{sphalerite}] \rightarrow \text{CaO}[\text{rocksalt}]$	−79	−78.9	exo	exo
$\text{CaN} \rightarrow \frac{1}{3}\text{Ca}_3\text{N}_2 + \frac{1}{6}\text{N}_2$	−205	−204.8	exo	exo
$\text{Mn}_2\text{O}_7 \rightarrow 2\text{MnO}_2 + \frac{3}{2}\text{O}_2$	−186	−187.7	exo	exo

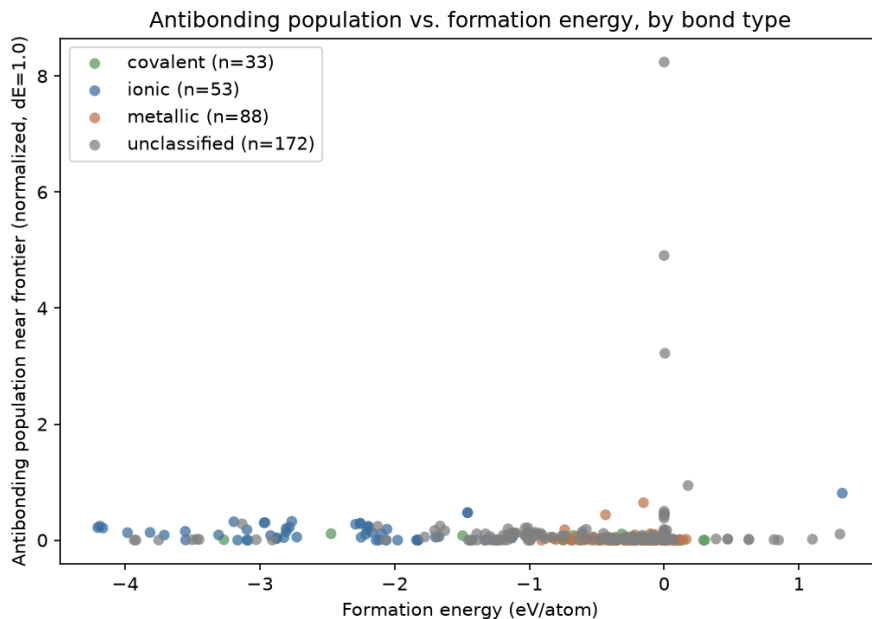
The Mn_2O_7 case is the manuscript’s own guard-rail example: exobondic by this static sign criterion, yet Mn_2O_7 is a real, observed compound that simply decomposes slowly (gradual O_2 loss) rather than by sudden bond rupture — `classify.py` structurally attaches a warning to *every* `UNSTABLE_NONEXISTENT` verdict for exactly this reason, rather than hardcoding Mn_2O_7 as a special case.

The full list of `extension4`’s 89 target compounds, its 16 elemental references, and the 89 case-1 decomposition reactions computed from them (with their own ΔICOHP /atom and endobondic/exobondic label, distinct from the 7 literature cases above) is given in Appendix E and Appendix F.

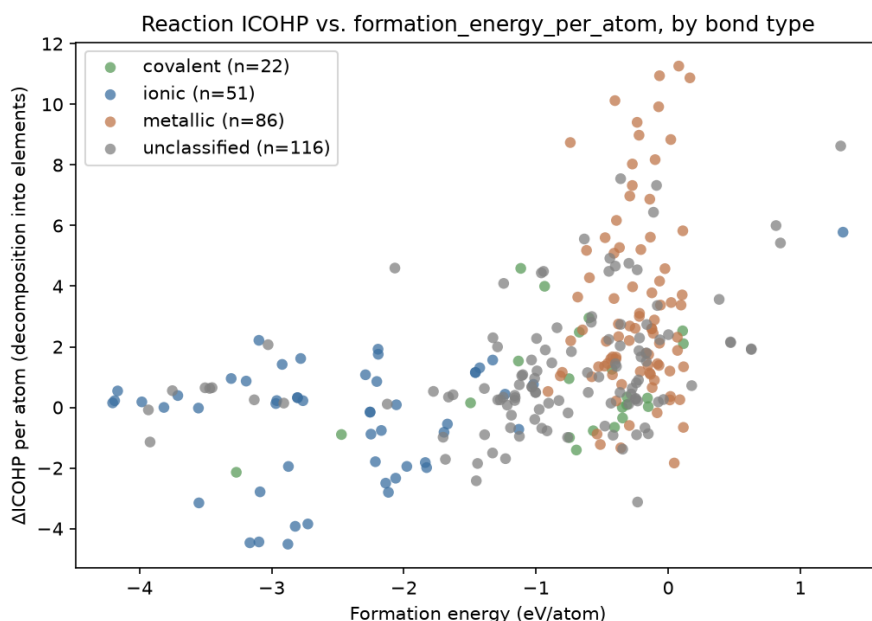
12.2 Missions #3–#5, reprocessed at 349 compounds

Every per-compound descriptor script was rerun unmodified over the grown dataset — `extension4`’s constituent elements were all already covered by the existing elemental-reference set, so no new element calculations were needed.

Mission #4 (antibonding): $\rho = -0.282$, $p = 9.2 \times 10^{-8}$ ($n = 346$) — weaker than the original $\rho = -0.328$, but the `bond_type=covalent` subgroup got *stronger* ($n = 33$, $\rho = -0.551$, $p = 0.0009$, vs. the original $\rho = -0.490$ at $n = 23$) and remains the single most robust stratified result in the project. `is_metal=False` newly survives ($n = 157$, $p = 0.001$). The old `bond_type=ionic` result ($n = 9$, $\rho = -0.683$) did not replicate once the sample grew to $n = 53$ ($\rho = -0.184$, not significant) — a clean demonstration of why $n < 15$ subgroups are flagged rather than trusted.



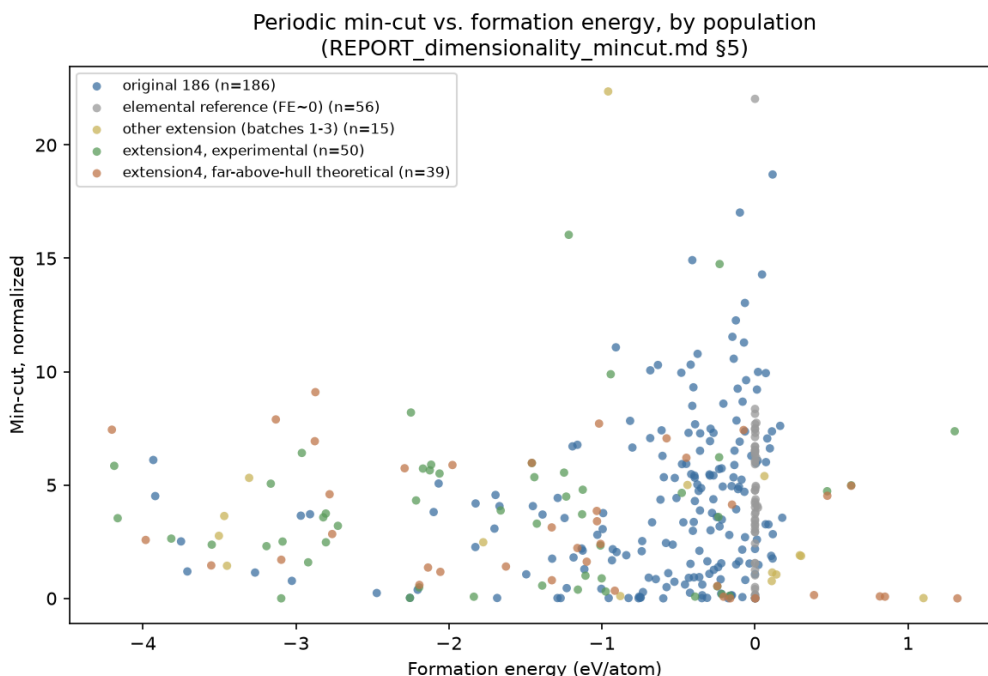
Mission #5 (reaction ICOHP): $\rho = 0.502$, $p = 6.3 \times 10^{-19}$ ($n = 275$) — up from $\rho = 0.370$, and now the clear strongest correlation in the project by a wider margin than before. `bond_type` stratification still fully explains that layer of the pooled signal (Kruskal-Wallis $p = 1.1 \times 10^{-15}$, no stratum survives) — **but `is_metal` stratification now survives on both sides** (`is_metal=False`: $\rho = 0.319$, $p = 0.0001$; `is_metal=True`: $\rho = 0.256$, $p = 0.0025$), where at $n = 186$ neither did. This revises the previous verdict that “the between-group confound explains the entirety of the observable signal” — it explains the `bond_type`-level pooling completely, but not a real, newly-detectable within-`is_metal`-group relationship. Case 2 (polymorph comparison), extended from 8 to **49 groups** specifically because `extension4` was designed to fix that sampling gap: 53.1% agreement between most-bonding and most-stable member, an exact Poisson-binomial test against the 45.9% chance baseline gives $P = 0.19$ — still no signal, but now a real test instead of an 8-point exploratory pass.



12.3 Why did the min-cut correlation collapse?

Mission #3’s headline, $\rho = 0.285$ at $n = 186$, collapsed to $\rho = 0.089$, $p = 0.098$ (not significant) at $n = 349$. **This is not a refutation of the original result** — isolated, the original 186 compounds still give $\rho = 0.2845$, $p = 8.3 \times 10^{-5}$, essentially identical to the mission-#3 number. The collapse is a dilution artifact from pooling populations added for unrelated reasons:

- 65 of the 74 extension-batch-1–3 compounds are single-element reference structures (added for mission #5’s elemental references), sitting at `formation_energy_per_atom` ≈ 0 by construction with widely-scattered min-cut values — mechanically dilutes a rank correlation regardless of any true relationship. Excluding them recovers $\rho = 0.2157$ ($n = 193$).
- `extension4` is not one population but two, with *opposite-sign* correlations: the experimental half is flat ($\rho = 0.06$, not significant, $n = 50$); the deliberately far-above-hull theoretical half shows a *significant* correlation of the *opposite* sign ($\rho = -0.40$, $p = 0.011$, $n = 39$) — not established here whether this reflects real physics of far-from-hull structures or a selection artifact of picking the single highest-EAH entry per formula.



Recommendation: any future min-cut-vs-formation-energy number (or, by the same logic, any other descriptor’s) should be conditioned on `family`/selection-provenance, the same way this project already conditions on `bond_type`/`is_metal` — pooling compounds selected for unrelated reasons is exactly the between-group confound this project’s own methodology exists to catch, and it had not been applied to the dataset’s own growing denominator until now.

Still no SISSO. Reaction ICOHP is now the strongest raw correlate of `formation_energy_per_atom` in the project, and the first descriptor whose signal is not fully explained away by between-group clustering at every level tested — the strongest argument yet for a stratified (`is_metal`-conditional) feature search over a pooled one, still not attempted.

Detailed reports: `analysis/REPORT_dimensionality_mincut.md`, `analysis/REPORT_antibonding.md`, `analysis/REPORT_reaction_icohp.md`.

A Compound List

The 186 compounds in the dataset, grouped by category (family), with their Materials Project identifier, PBE thermodynamic hull distance (`energy_above_hull`), and the minimum ICOHP percolation weight (`icohp_percolation_weight_min`, this work’s main metric). Auto-generated from `analysis/percolation_vs_hull.csv` by `report/gen_appendix.py`.

Table 1: Complete list of the 186 compounds, grouped by category, with their Materials Project identifier, PBE hull distance, and minimum ICOHP percolation weight.

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
Experimental, stable (hull) ($n = 63$)				
As ₂ S ₃	mp-1189572	covalent	0.0000	1.00e-05
AsF ₃	mp-28027	covalent	0.0000	0.00e+00
BI ₃	mp-23189	covalent	0.0000	1.00e-05
BW	mp-7832	—	0.0000	1.90e-04
BaAu ₂	mp-30363	metallic	0.0000	5.19e-02
BaCl ₂	mp-568662	ionic	0.0000	1.31e-02
BaH ₂	mp-23715	—	0.0000	2.40e-04
Be ₂ Cu	mp-2031	metallic	0.0000	1.15e-03
Be ₂ Nb ₃	mp-11272	metallic	0.0000	6.36e-03
Be ₂ V	mp-11281	metallic	0.0000	6.80e-04
CaAl ₄	mp-1749	metallic	0.0000	1.03e-02
CaSn	mp-2450	metallic	0.0000	5.23e-03
CdPd	mp-1696	metallic	0.0000	9.00e-05
CoB	mp-20857	—	0.0000	6.00e-05
Cs ₂ As ₃	mp-15557	—	0.0000	2.02e-02
FeS	mp-505531	—	0.0000	3.05e-03
Ge ₂ Os	mp-10032	metallic	0.0000	1.21e-02
HfTc ₂	mp-1095669	metallic	0.0000	1.77e-02
Hg ₄ Pt	mp-936	metallic	0.0000	3.36e-03
HgF ₂	mp-8177	—	0.0000	8.95e-03
InBr	mp-22870	covalent	0.0000	1.02e-02
KI	mp-22898	ionic	0.0000	2.26e-02
KN ₃	mp-827	—	0.0000	2.80e-04
Li ₂ Se	mp-2286	—	0.0000	1.16e-02
LiB	mp-1001835	—	0.0000	2.91e-03
Mg ₂ Ni	mp-2137	metallic	0.0000	1.60e-04
MgP ₄	mp-384	—	0.0000	0.00e+00
MnAl ₁₂	mp-1104165	metallic	0.0000	1.42e-02
MoSe ₂	mp-1634	covalent	0.0000	1.16e-03
NaS	mp-2400	—	0.0000	0.00e+00
Na	mp-10172	—	0.0000	1.97e-03
Nb ₃ Ir	mp-1458	metallic	0.0000	9.02e-03
NbGa ₃	mp-1973	metallic	0.0000	8.58e-03
PI ₂	mp-29443	covalent	0.0000	2.30e-04
PbSe	mp-2201	covalent	0.0000	1.66e-02
RbI	mp-22903	ionic	0.0000	2.00e-02
Sc ₅ Ge ₃	mp-17190	metallic	0.0000	5.30e-04
ScAg ₂	mp-1018128	metallic	0.0000	1.70e-04
ScTe	mp-10026	—	0.0000	7.12e-03
SnPt	mp-19856	metallic	0.0000	1.29e-02
SrO ₂	mp-2697	ionic	0.0000	4.40e-04
SrSi	mp-2661	metallic	0.0000	7.00e-05
Ta ₅ Ge ₃	mp-17593	metallic	0.0000	3.46e-02

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
TaP	mp-1067587	—	0.0000	3.14e-02
TaSb2	mp-11697	metallic	0.0000	1.17e-02
Te2Pd	mp-782	—	0.0000	1.44e-02
Te2Ru	mp-267	covalent	0.0000	4.34e-03
TeO2	mp-8377	covalent	0.0000	4.00e-05
Ti3In	mp-866046	metallic	0.0000	5.01e-03
Ti3Pt	mp-2339	metallic	0.0000	1.78e-03
TiBe12	mp-1104067	metallic	0.0000	2.40e-04
TiNi	mp-1048	metallic	0.0000	3.91e-03
TiO	mp-1071163	—	0.0000	6.40e-04
TiSi2	mp-1077503	metallic	0.0000	6.18e-03
TiCl	mp-571079	—	0.0000	7.17e-03
VNi2	mp-11531	metallic	0.0000	3.88e-03
WO2	mp-1524394	—	0.0000	2.20e-04
Zr2Au	mp-2348	metallic	0.0000	3.06e-02
Zr5Pb3	mp-681992	metallic	0.0000	2.00e-05
ZrMo2	mp-2049	metallic	0.0000	1.05e-02
Si	mp-149	covalent	0.0000	3.17e-03
NaCl	mp-22862	ionic	0.0000	3.07e-02
AlNi	mp-1487	metallic	0.0000	4.19e-03
Experimental, metastable ($n = 63$)				
CdI2	mp-567259	—	0.0005	2.38e-02
MgCl2	mp-570259	ionic	0.0006	1.90e-04
C	mp-169	covalent	0.0008	2.50e-04
ZrBr	mp-1065846	—	0.0008	7.88e-03
Li3Hg	mp-1646	metallic	0.0013	1.20e-03
Ta7Co6	mp-1104548	metallic	0.0015	3.10e-04
CaGa4	mp-2461	metallic	0.0016	1.91e-02
InCl	mp-571636	covalent	0.0016	8.50e-03
NiB	mp-14019	—	0.0018	4.10e-04
MgH2	mp-23711	—	0.0019	1.00e-05
Sn7Ir5	mp-20466	metallic	0.0024	1.00e-02
NaN3	mp-1066400	—	0.0029	7.00e-04
ClF3	mp-556767	—	0.0030	0.00e+00
SiS2	mp-1095294	covalent	0.0030	1.00e-04
Sr5Sn3	mp-17720	metallic	0.0030	1.08e-02
Te2Ir	mp-1551	—	0.0030	2.80e-03
Li13Sn5	mp-30769	metallic	0.0033	8.50e-04
H3N	mp-643432	covalent	0.0038	0.00e+00
Cs2Se	mp-1011696	—	0.0042	2.62e-02
In3Ir	mp-630976	metallic	0.0043	2.04e-03
HI	mp-697025	—	0.0044	0.00e+00
Rb2Hg7	mp-31474	metallic	0.0045	3.14e-02
PbAu2	mp-19871	metallic	0.0047	2.01e-03
BeCu	mp-2323	metallic	0.0050	4.80e-04
SiO2	mp-546794	covalent	0.0056	1.50e-04
IF7	mp-27988	—	0.0063	0.00e+00
K	mp-10157	—	0.0069	8.07e-02
Al3Os2	mp-16521	metallic	0.0074	6.23e-03
Nb2O5	mp-1595	—	0.0075	1.00e-04
LiBr	mp-23259	ionic	0.0099	9.88e-02
ZnAg	mp-1912	metallic	0.0112	9.67e-03
MgPd3	mp-7753	metallic	0.0114	2.21e-03
AsPd2	mp-1080831	—	0.0118	3.50e-04

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
GeSe2	mp-10074	covalent	0.0125	4.00e-05
Re3Ge7	mp-29141	metallic	0.0126	2.42e-03
Cu3Ge	mp-19724	metallic	0.0126	5.30e-04
Ti3Ge	mp-1189044	metallic	0.0130	1.40e-02
ZrGa3	mp-30685	metallic	0.0130	7.64e-03
SrAl2	mp-1638	metallic	0.0136	5.36e-03
NiS	mp-1547	—	0.0142	3.00e-04
Al12W	mp-11227	metallic	0.0152	1.37e-02
CsH	mp-632319	—	0.0161	2.66e-02
CaGa2	mp-992	metallic	0.0170	1.66e-02
NiHg4	mp-570835	metallic	0.0171	1.09e-03
SiRh	mp-2287	metallic	0.0246	1.10e-02
BaSn5	mp-571353	metallic	0.0262	4.58e-03
PS	mp-612	covalent	0.0265	8.00e-05
AsRh	mp-22079	—	0.0269	1.93e-03
Te2Au	mp-567525	—	0.0274	1.62e-02
Be3N2	mp-6977	—	0.0315	2.00e-05
Y2O3	mp-558573	—	0.0342	5.00e-05
AgO	mp-1079720	—	0.0423	4.00e-05
FeTe2	mp-1102002	—	0.0449	5.20e-04
H2O	mp-2739191	—	0.0494	0.00e+00
CdO2	mp-2310	—	0.0562	9.00e-05
Mo3Se4	mp-21021	—	0.0609	1.84e-03
PbSe	mp-22009	covalent	0.0731	1.68e-02
Be12Pd	mp-12666	metallic	0.0795	5.00e-05
Sr2As	mp-15697	—	0.0800	2.06e-03
AsRh2	mp-1102364	—	0.0806	4.63e-03
ZrRh	mp-2808	metallic	0.0839	3.50e-02
K2S	mp-559278	—	0.0873	1.07e-02
Ge4Rh	mp-1104286	metallic	0.0907	2.08e-03
Theoretical, metastable ($n = 60$)				
Li3Ag	mp-976408	metallic	0.0013	1.15e-02
Y3Co	mp-1189329	metallic	0.0016	1.50e-02
TiNi	mp-1067248	metallic	0.0020	5.29e-03
TaP	mp-1187244	—	0.0033	3.59e-02
Sr3As4	mp-678007	—	0.0062	4.47e-03
Ga2Se3	mp-1224809	covalent	0.0063	1.14e-02
MnN	mp-1009130	covalent	0.0151	1.85e-03
BeCl2	mp-1190080	ionic	0.0187	2.37e-03
MgSn	mp-1094801	metallic	0.0212	4.26e-03
TiW	mp-1216621	metallic	0.0225	2.25e-03
SrTi3	mp-1206636	metallic	0.0294	3.29e-03
IrCl3	mp-729507	—	0.0301	2.72e-03
Ta4C3	mp-1218000	—	0.0335	1.44e-03
Sc2O3	mp-755313	—	0.0364	5.10e-04
SnBr2	mp-978985	covalent	0.0405	3.40e-04
Sn7Pd5	mp-1219006	metallic	0.0414	2.30e-03
Tl4Sn	mp-1216560	metallic	0.0459	1.00e+00
LiTi3	mp-1185253	metallic	0.0467	3.35e-03
PtCl2	mp-1219748	—	0.0490	5.04e-03
HCl	mp-684609	—	0.0536	2.50e-04
MgSn	mp-1094669	metallic	0.0560	2.60e-04
K5As4	mp-684799	—	0.0564	8.80e-03
KSr3	mp-1185151	metallic	0.0573	2.65e-02

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
NbS2	mp-774873	—	0.0576	1.20e-04
YAg3	mp-1187811	metallic	0.0594	1.12e-03
B3Ir2	mp-1228647	—	0.0609	2.80e-04
TiMo2	mp-1216675	metallic	0.0613	3.77e-02
Cr2C	mp-1226378	—	0.0625	2.08e-03
ZrO2	mp-775909	—	0.0627	1.20e-04
Zn	mp-2647117	—	0.0717	1.04e-02
IN2	mp-1097011	—	0.0730	4.70e-04
CoSe2	mp-1390196	—	0.0734	8.40e-04
GeS	mp-12910	covalent	0.0762	2.26e-03
ZrTi	mp-1215236	metallic	0.0883	3.15e-02
VCo3	mp-1095057	metallic	0.0942	2.93e-03
Tl3Ga	mp-1187539	metallic	0.0972	5.00e-05
CaGe2	mp-643542	metallic	0.0979	1.30e-02
NbRh2	mp-1220414	metallic	0.1043	1.76e-02
TcW	mp-1217337	metallic	0.1056	4.10e-03
MgF2	mp-560236	ionic	0.1076	2.00e-05
VN	mp-1017532	—	0.1080	4.00e-05
Na2Cl	mp-990084	—	0.1105	3.76e-02
ZnPb3	mp-1187949	metallic	0.1107	4.51e-03
Sr3Sn	mp-1187136	metallic	0.1111	7.94e-03
TaRe	mp-1217894	metallic	0.1113	6.98e-02
Al4Si	mp-1228120	metallic	0.1149	8.54e-02
B6Pb	mp-1096825	covalent	0.1151	2.35e-03
ScTc3	mp-867262	metallic	0.1154	3.31e-02
CrMo	mp-1226200	metallic	0.1163	4.23e-02
BPd5	mp-1228665	—	0.1325	2.42e-02
Zr3Be	mp-1188016	metallic	0.1430	1.05e-03
Na3Cl	mp-1064484	—	0.1456	2.55e-03
SiAu3	mp-1186998	metallic	0.1643	5.68e-03
InSe2	mp-1232036	—	0.1733	1.64e-03
TiSi2	mp-570991	metallic	0.1796	1.35e-02
BeAu3	mp-983539	metallic	0.1818	2.41e-02
Zr3Ge	mp-1188039	metallic	0.1851	6.59e-03
AlSb	mp-1023918	metallic	0.1878	3.60e-04
Mg15B	mp-1023515	—	0.1943	2.80e-04
HfMo	mp-1224314	metallic	0.1991	2.00e-02

B VASP Calculation Parameters

Fixed INCAR parameters, identical across all 186 compounds (static, LOBSTER-compatible run) — only NBANDS (appendix D) and the k-point grid vary by compound.

Parameter	Value	Description
PREC	Accurate	Overall precision (charge grid, projectors)
ENCUT	600.0	Plane-wave cutoff energy (eV)
EDIFF	1e-06	Electronic convergence criterion (eV)
IBRION	-1	Ionic dynamics type (-1 = static, no relaxation)
NSW	0	Number of ionic steps (0 = static run on the MP structure)
ISMear	0	Smearing type (0 = Gaussian)
SIGMA	0.05	Smearing width (eV)
LASPH	True	Non-spherical corrections to the PAW potential
ISYM	-1	Symmetry (-1 = disabled, required by LOBSTER)
LWAVE	True	Write WAVECAR (required by LOBSTER)
LCHARG	True	Write CHGCAR
NPAR	4	Band-level parallelization
KPAR	2	K-point-level parallelization

C PAW-PBE Potentials

One PAW-PBE potential per element present in the dataset (pymatgen’s `MPRelaxSet` library — the same one Materials Project used to relax the input structures), with one documented exception for tungsten (§7): the recommended `W_pv` variant is absent from the local mirror, and `W_sv` turned out to be corrupted (missing `LEXCH` field) — plain `W` is used instead, verified valid.

Table 2: PAW-PBE potentials used, one per element present in the dataset (pymatgen’s `MPRelaxSet` library, with one documented exception: `W` substituted for `W_pv`/`W_sv`, see §7).

Element	PAW-PBE potential
Ag	Ag (02Apr2005)
Al	Al (04Jan2001)
As	As (22Sep2009)
Au	Au (04Oct2007)
B	B (06Sep2000)
Ba	Ba_sv (06Sep2000)
Be	Be_sv (06Sep2000)
Br	Br (06Sep2000)
C	C (08Apr2002)
Ca	Ca_sv (06Sep2000)
Cd	Cd (06Sep2000)
Cl	Cl (06Sep2000)
Co	Co (02Aug2007)
Cr	Cr_pv (02Aug2007)
Cs	Cs_sv (25Jan2019)
Cu	Cu_pv (06Sep2000)
F	F (08Apr2002)
Fe	Fe_pv (02Aug2007)
Ga	Ga_d (06Jul2010)
Ge	Ge_d (03Jul2007)
H	H (15Jun2001)
Hf	Hf_pv (06Sep2000)

Element	PAW-PBE potential
Hg	Hg (06Sep2000)
I	I (08Apr2002)
In	In_d (06Sep2000)
Ir	Ir (06Sep2000)
K	K_sv (06Sep2000)
Li	Li_sv (10Sep2004)
Mg	Mg_pv (13Apr2007)
Mn	Mn_pv (02Aug2007)
Mo	Mo_pv (04Feb2005)
N	N (08Apr2002)
Na	Na_pv (19Sep2006)
Nb	Nb_pv (08Apr2002)
Ni	Ni_pv (06Sep2000)
O	O (08Apr2002)
Os	Os_pv (20Jan2003)
P	P (06Sep2000)
Pb	Pb_d (06Sep2000)
Pd	Pd (04Jan2005)
Pt	Pt (04Feb2005)
Rb	Rb_sv (06Sep2000)
Re	Re_pv (06Sep2000)
Rh	Rh_pv (25Jan2005)
Ru	Ru_pv (28Jan2005)
S	S (06Sep2000)
Sb	Sb (06Sep2000)
Sc	Sc_sv (07Sep2000)
Se	Se (06Sep2000)
Si	Si (05Jan2001)
Sn	Sn_d (06Sep2000)
Sr	Sr_sv (07Sep2000)
Ta	Ta_pv (07Sep2000)
Tc	Tc_pv (04Feb2005)
Te	Te (08Apr2002)
Ti	Ti_pv (07Sep2000)
Tl	Tl_d (06Sep2000)
V	V_pv (07Sep2000)
W	W (08Apr2002)
Y	Y_sv (25May2007)
Zn	Zn (06Sep2000)
Zr	Zr_sv (04Jan2005)

D k-point Grid and NBANDS per Compound

k-point grid auto-generated by `pymatgen.io.vasp.Kpoints.automatic_density_by_vol` (density `kppv1=100`, Γ -centered or Monkhorst–Pack mesh depending on cell symmetry) and `NBANDS`, derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`) — a strictly sufficient value for the local-basis projection, not a fixed guess.

Table 3: k-point grid (pymatgen automatic density, `kppvol=100`) and number of bands (derived from the LOBSTER basis) per compound.

Formula	mp-id	k-point grid	NBANDS
Experimental, stable (hull) ($n = 63$)			
As ₂ S ₃	mp-1189572	5 3 2 (G)	80
AsF ₃	mp-28027	5 4 4 (G)	64
BI ₃	mp-23189	4 4 4 (G)	32
BW	mp-7832	9 9 3 (G)	52
BaAu ₂	mp-30363	6 6 7 (G)	23
BaCl ₂	mp-568662	6 6 6 (G)	13
BaH ₂	mp-23715	6 4 3 (G)	28
Be ₂ Cu	mp-2031	7 7 7 (G)	38
Be ₂ Nb ₃	mp-11272	8 4 4 (M)	74
Be ₂ V	mp-11281	7 7 4 (G)	76
CaAl ₄	mp-1749	7 7 4 (G)	21
CaSn	mp-2450	6 6 4 (M)	28
CdPd	mp-1696	7 6 6 (G)	36
CoB	mp-20857	9 7 5 (G)	52
Cs ₂ As ₃	mp-15557	4 4 3 (G)	44
FeS	mp-505531	8 8 5 (G)	26
Ge ₂ Os	mp-10032	9 6 4 (G)	54
HfTc ₂	mp-1095669	5 5 3 (G)	108
Hg ₄ Pt	mp-936	5 5 5 (G)	45
HgF ₂	mp-8177	8 8 8 (G)	17
InBr	mp-22870	6 6 4 (M)	26
KI	mp-22898	6 6 6 (G)	9
KN ₃	mp-827	5 5 5 (G)	34
Li ₂ Se	mp-2286	7 7 7 (G)	14
LiB	mp-1001835	7 7 9 (G)	18
Mg ₂ Ni	mp-2137	5 5 2 (G)	102
MgP ₄	mp-384	5 5 3 (G)	40
MnAl ₁₂	mp-1104165	4 4 4 (M)	57
MoSe ₂	mp-1634	9 9 2 (G)	34
NaS	mp-2400	6 6 3 (G)	32
Na	mp-10172	8 8 4 (G)	8
Nb ₃ Ir	mp-1458	5 5 5 (G)	72
NbGa ₃	mp-1973	8 8 6 (M)	36
PI ₂	mp-29443	6 4 3 (G)	24
PbSe	mp-2201	7 7 7 (G)	13
RbI	mp-22903	6 6 6 (G)	9
Sc ₅ Ge ₃	mp-17190	3 3 5 (G)	154
ScAg ₂	mp-1018128	8 8 5 (G)	28
ScTe	mp-10026	7 7 4 (G)	28
SnPt	mp-19856	7 7 5 (G)	36
SrO ₂	mp-2697	8 8 7 (G)	13
SrSi	mp-2661	7 6 4 (G)	18
Ta ₅ Ge ₃	mp-17593	4 4 4 (M)	144
TaP	mp-1067587	9 9 4 (G)	26
TaSb ₂	mp-11697	8 5 4 (G)	34
Te ₂ Pd	mp-782	7 7 5 (G)	17
Te ₂ Ru	mp-267	7 5 4 (G)	34
TeO ₂	mp-8377	6 5 3 (G)	48
Ti ₃ In	mp-866046	5 5 6 (G)	72
Ti ₃ Pt	mp-2339	5 5 5 (G)	72
TiBe ₁₂	mp-1104067	7 5 5 (G)	69

Table 3 (continued)

Formula	mp-id	k-point grid	NBANDS
TiNi	mp-1048	10 7 6 (G)	36
TiO	mp-1071163	10 6 6 (G)	39
TiSi2	mp-1077503	8 8 4 (M)	34
TlCl	mp-571079	6 6 4 (M)	26
VNi2	mp-11531	12 8 7 (G)	27
WO2	mp-1524394	6 5 5 (G)	68
Zr2Au	mp-2348	9 9 4 (G)	29
Zr5Pb3	mp-681992	3 3 5 (G)	154
ZrMo2	mp-2049	6 6 6 (G)	56
Si	mp-149	8 8 8 (G)	100
NaCl	mp-22862	8 8 8 (G)	100
AlNi	mp-1487	10 10 10 (M)	100
Experimental, metastable ($n = 63$)			
CdI2	mp-567259	7 7 4 (G)	17
MgCl2	mp-570259	8 8 5 (G)	12
C	mp-169	12 12 8 (M)	100
ZrBr	mp-1065846	8 8 3 (G)	28
Li3Hg	mp-1646	7 7 7 (G)	24
Ta7Co6	mp-1104548	6 6 3 (G)	117
CaGa4	mp-2461	7 7 5 (G)	41
InCl	mp-571636	6 6 4 (M)	26
NiB	mp-14019	10 10 7 (G)	26
MgH2	mp-23711	6 5 5 (G)	24
Sn7Ir5	mp-20466	4 4 4 (M)	108
NaN3	mp-1066400	7 7 7 (G)	16
ClF3	mp-556767	6 4 3 (G)	64
SiS2	mp-1095294	5 4 3 (G)	48
Sr5Sn3	mp-17720	3 3 3 (G)	104
Te2Ir	mp-1551	7 5 2 (G)	51
Li13Sn5	mp-30769	6 6 1 (G)	110
H3N	mp-643432	8 5 5 (G)	28
Cs2Se	mp-1011696	5 3 2 (G)	56
In3Ir	mp-630976	4 4 4 (M)	144
HI	mp-697025	3 3 3 (G)	40
Rb2Hg7	mp-31474	4 4 4 (G)	73
PbAu2	mp-19871	5 5 5 (G)	54
BeCu	mp-2323	10 10 10 (M)	100
SiO2	mp-546794	6 6 6 (M)	24
IF7	mp-27988	5 5 3 (G)	64
K	mp-10157	6 6 6 (G)	5
Al3Os2	mp-16521	9 9 3 (G)	30
Nb2O5	mp-1595	7 7 4 (G)	38
LiBr	mp-23259	8 8 8 (G)	100
ZnAg	mp-1912	9 9 9 (G)	18
MgPd3	mp-7753	7 7 7 (G)	31
AsPd2	mp-1080831	8 4 4 (G)	66
GeSe2	mp-10074	5 5 5 (G)	34
Re3Ge7	mp-29141	9 6 1 (G)	180
Cu3Ge	mp-19724	6 6 5 (G)	72
Ti3Ge	mp-1189044	4 4 4 (M)	144
ZrGa3	mp-30685	7 7 5 (G)	37
SrAl2	mp-1638	5 5 5 (G)	26
NiS	mp-1547	6 6 6 (M)	39
Al12W	mp-11227	4 4 4 (M)	57

Table 3 (continued)

Formula	mp-id	k-point grid	NBANDS
CsH	mp-632319	7 7 7 (G)	6
CaGa2	mp-992	7 7 6 (G)	23
NiHg4	mp-570835	6 6 6 (M)	45
SiRh	mp-2287	6 6 5 (G)	52
BaSn5	mp-571353	5 5 4 (G)	50
PS	mp-612	4 4 3 (G)	64
AsRh	mp-22079	8 5 4 (G)	52
Te2Au	mp-567525	7 7 5 (G)	17
Be3N2	mp-6977	10 10 3 (G)	46
Y2O3	mp-558573	8 4 3 (G)	96
AgO	mp-1079720	8 5 5 (G)	52
FeTe2	mp-1102002	4 4 4 (M)	68
H2O	mp-2739191	7 7 4 (G)	24
CdO2	mp-2310	5 5 5 (G)	68
Mo3Se4	mp-21021	4 4 4 (M)	86
PbSe	mp-22009	6 6 2 (G)	26
Be12Pd	mp-12666	6 6 6 (M)	69
Sr2As	mp-15697	6 6 3 (G)	28
AsRh2	mp-1102364	7 4 3 (G)	88
ZrRh	mp-2808	8 8 8 (M)	19
K2S	mp-559278	5 5 4 (G)	28
Ge4Rh	mp-1104286	4 4 3 (G)	135
Theoretical, metastable ($n = 60$)			
Li3Ag	mp-976408	7 7 7 (G)	24
Y3Co	mp-1189329	3 4 4 (G)	156
TiNi	mp-1067248	10 7 6 (G)	36
TaP	mp-1187244	9 9 9 (G)	13
Sr3As4	mp-678007	3 3 4 (G)	62
Ga2Se3	mp-1224809	5 5 5 (G)	30
MnN	mp-1009130	10 10 10 (G)	13
BeCl2	mp-1190080	3 3 3 (G)	78
MgSn	mp-1094801	8 8 6 (M)	13
TiW	mp-1216621	10 10 6 (M)	18
SrTi3	mp-1206636	5 5 5 (G)	32
IrCl3	mp-729507	3 3 6 (G)	84
Ta4C3	mp-1218000	6 6 6 (M)	48
Sc2O3	mp-755313	6 5 6 (G)	64
SnBr2	mp-978985	4 4 6 (M)	34
Sn7Pd5	mp-1219006	5 5 3 (G)	108
Tl4Sn	mp-1216560	5 5 5 (G)	45
LiTi3	mp-1185253	6 6 6 (M)	32
PtCl2	mp-1219748	5 5 4 (G)	34
HCl	mp-684609	7 7 7 (G)	5
MgSn	mp-1094669	8 8 8 (M)	13
K5As4	mp-684799	4 4 3 (G)	41
KSn3	mp-1185151	4 4 5 (G)	64
NbS2	mp-774873	10 6 2 (M)	34
YAg3	mp-1187811	6 6 6 (G)	37
B3Ir2	mp-1228647	9 5 4 (G)	60
TiMo2	mp-1216675	5 6 14 (G)	27
Cr2C	mp-1226378	10 10 7 (G)	22
ZrO2	mp-775909	10 6 4 (M)	36
Zn	mp-2647117	11 11 11 (G)	9
IN2	mp-1097011	5 5 5 (G)	24

Table 3 (continued)

Formula	mp-id	k-point grid	NBANDS
CoSe2	mp-1390196	4 4 4 (G)	68
GeS	mp-12910	7 7 5 (G)	26
ZrTi	mp-1215236	9 9 6 (G)	19
VCo3	mp-1095057	7 6 6 (G)	72
Tl3Ga	mp-1187539	5 5 5 (G)	36
CaGe2	mp-643542	7 7 6 (G)	23
NbRh2	mp-1220414	10 6 4 (M)	54
TcW	mp-1217337	10 10 6 (M)	18
MgF2	mp-560236	6 6 5 (G)	24
VN	mp-1017532	11 11 5 (G)	26
Na2Cl	mp-990084	6 6 6 (M)	12
ZnPb3	mp-1187949	5 5 5 (G)	36
Sr3Sn	mp-1187136	5 5 5 (G)	24
TaRe	mp-1217894	10 10 6 (M)	18
Al4Si	mp-1228120	6 6 6 (M)	20
B6Pb	mp-1096825	6 6 6 (M)	33
ScTc3	mp-867262	7 7 7 (G)	37
CrMo	mp-1226200	11 11 7 (G)	18
BPd5	mp-1228665	6 6 6 (M)	49
Zr3Be	mp-1188016	6 6 6 (M)	35
Na3Cl	mp-1064484	8 8 3 (G)	16
SiAu3	mp-1186998	7 7 7 (G)	31
InSe2	mp-1232036	7 7 3 (G)	34
TiSi2	mp-570991	8 8 8 (M)	17
BeAu3	mp-983539	7 7 7 (G)	32
Zr3Ge	mp-1188039	4 4 6 (G)	78
AlSb	mp-1023918	6 6 5 (G)	16
Mg15B	mp-1023515	4 4 3 (G)	64
HfMo	mp-1224314	10 10 6 (M)	18

E extension4: Compounds and Elements

Every compound computed in the **extension4** campaign (§12): 89 target compounds (alkali/alkaline-earth binaries against N/O/F/P/S/Cl, half experimentally known, half deliberately far above the convex hull) and the 16 elemental reference structures used to balance their case-1 decomposition reactions (Appendix F). Auto-generated from `mp_dataset/extension4_candidates.json` and each compound’s `mp_metadata.json` by `report/gen_appendix_extension4.py`.

Table 4: Target compounds of the **extension4** campaign (89) and elemental references used for their decomposition (16) – `mp_dataset/download_extension4.py` / `ELEMENT_REFERENCE`.

Formula	mp-id	Type	Space group	E_{hull} (eV/at)	Role
Ba	mp-122	exp.	Im-3m	0.0000	element
Be	mp-87	exp.	P6_3/mmc	0.0000	element
Ca	mp-21	exp.	Im-3m	0.0056	element
Cl2	mp-22848	exp.	Cmce	0.0000	element
Cs	mp-1	exp.	Im-3m	0.0193	element
F2	mp-561203	exp.	C2/c	0.0031	element
K	mp-10157	exp.	Fm-3m	0.0069	element
Li	mp-1018134	exp.	R-3m	0.0000	element
Mg	mp-153	exp.	P6_3/mmc	0.0000	element

Formula	mp-id	Type	Space group	E_{hull} (eV/at)	Role
N2	mp-1059834	theo.	Imma	1.1010	element
Na	mp-10172	exp.	P6 ₃ /mmc	0.0000	element
O2	mp-1524462	exp.	C2/m	0.0000	element
P	mp-568348	exp.	P2/c	0.0000	element
Rb	mp-70	exp.	Im-3m	0.0092	element
S	mp-77	exp.	Fddd	0.0000	element
Sr	mp-139	exp.	P6 ₃ /mmc	0.0000	element
Ba5P4	mp-17566	exp.	Pnma	0.0000	target
BaCl2	mp-23199	exp.	Pnma	0.0068	target
BaF3	mp-1080821	exp.	P2 ₁ /m	0.0000	target
BaF3	mp-1183303	theo.	I4/mmm	0.0612	target
BaN2	mp-1102371	exp.	Cc	2.0568	target
BaO	mp-1008500	exp.	P6 ₃ /mmc	0.0189	target
BaO	mp-1950989	theo.	P6 ₃ /mmc	0.0434	target
BaS3	mp-556296	exp.	P2 ₁₂ ₁₂	0.0436	target
Be3N2	mp-18337	exp.	Ia-3	0.0000	target
Be3N2	mp-1017550	theo.	P-3m1	0.2081	target
Be3P2	mp-567841	exp.	Ia-3	0.0000	target
Be3P2	mp-1084771	theo.	Pn-3m	0.1567	target
BeCl2	mp-23267	exp.	Ibam	0.0087	target
BeCl2	mp-1172909	theo.	I-43m	3.1688	target
BeF2	mp-15951	exp.	P3 ₁₂₁	0.0004	target
BeF2	mp-975644	theo.	P6/mmm	2.5181	target
BeO	mp-7599	exp.	P4 ₂ /mmn	0.0153	target
BeO	mp-1778	theo.	F-43m	0.0070	target
Ca3N2	mp-1047	exp.	R-3c	0.0167	target
Ca3N2	mp-1013524	theo.	Pm-3m	0.5112	target
CaCl2	mp-23214	exp.	Pnnm	0.0011	target
CaCl2	mp-1120739	theo.	Fm-3m	0.0422	target
CaF2	mp-10464	exp.	Pnma	0.0346	target
CaF2	mp-554355	theo.	P4/mmm	0.2403	target
CaO	mp-1079707	exp.	Cmce	0.2098	target
CaO	mp-1181903	theo.	Fd-3m	2.7281	target
Cs2S	mp-1077814	exp.	P6 ₃ /mmc	0.0667	target
CsCl	mp-573697	exp.	Fm-3m	0.0071	target
CsF	mp-8455	exp.	Pm-3m	0.1088	target
CsN3	mp-510557	exp.	I4/mcm	0.0000	target
CsN3	mp-1225892	theo.	P4/mmm	0.0113	target
CsO2	mp-1441	exp.	I4/mmm	0.0349	target
CsO2	mp-1096936	theo.	Cmmm	0.0940	target
CsP7	mp-1201790	exp.	Pbcm	0.0000	target
CsP7	mp-1213206	theo.	Pca2 ₁	0.0001	target
K2O	mp-971	exp.	Fm-3m	0.0000	target
K2O	mp-1223784	theo.	P4/mmm	0.2000	target
K2S	mp-1022	exp.	Fm-3m	0.0000	target
KCl	mp-23289	exp.	Pm-3m	0.0776	target
KF3	mp-1544557	exp.	P2 ₁₂ ₁₂ ₁	0.0125	target
KF3	mp-2749823	theo.	P2 ₁	0.0119	target
KN3	mp-636056	theo.	I4/mcm	1.0370	target
KP	mp-1058315	exp.	R-3m	1.0080	target
KP	mp-1057787	theo.	Immm	1.0092	target
Li2O	mp-1960	exp.	Fm-3m	0.0000	target
Li2O	mp-755894	theo.	Pnma	0.0844	target
Li2S	mp-1125	exp.	Pnma	0.0615	target
Li2S	mp-557142	theo.	P6 ₃ /mmc	0.1750	target
Li3N	mp-2341	exp.	P6 ₃ /mmc	0.0039	target

Formula	mp-id	Type	Space group	E_{hull} (eV/at)	Role
LiCl	mp-22905	exp.	Fm-3m	0.0243	target
LiCl	mp-1185319	theo.	P6_3mc	0.0039	target
LiF	mp-1138	exp.	Fm-3m	0.0000	target
LiF	mp-1009009	theo.	Pm-3m	0.2875	target
LiP5	mp-2412	exp.	Pna2_1	0.0077	target
LiP5	mp-32760	theo.	Pna2_1	0.0963	target
MgCl2	mp-23210	exp.	R-3m	0.0000	target
MgCl2	mp-569626	theo.	Ama2	0.1991	target
MgF2	mp-1072956	exp.	Pnnm	0.0025	target
MgF2	mp-1062842	theo.	Amm2	0.2636	target
MgO	mp-549706	exp.	P6_3mc	0.1180	target
MgO	mp-1009127	theo.	Pm-3m	0.7484	target
MgS	mp-1315	exp.	Fm-3m	0.0000	target
MgS	mp-13032	theo.	F-43m	0.0343	target
Na2Cl	mp-1077015	theo.	Cmmm	0.2391	target
Na2O	mp-2352	exp.	Fm-3m	0.0000	target
Na2O	mp-1975297	theo.	C2/m	0.0981	target
NaN3	mp-22003	exp.	R-3m	0.0017	target
NaN3	mp-1065265	theo.	C2/m	1.0162	target
NaS	mp-409	exp.	P-62m	0.0023	target
NaS	mp-1180106	theo.	C2/m	1.0061	target
Rb2O	mp-1394	exp.	Fm-3m	0.0032	target
Rb2O	mp-2023482	theo.	P-3m1	0.0479	target
RbCl	mp-23299	exp.	Pm-3m	0.0298	target
RbF	mp-2064	exp.	Pm-3m	0.0516	target
RbF	mp-11718	theo.	Fm-3m	0.0000	target
RbN3	mp-581833	exp.	P4/mmm	0.0193	target
RbN3	mp-1219567	theo.	Amm2	0.7966	target
RbP	mp-1179688	exp.	C2/m	0.9879	target
RbP	mp-1058499	theo.	Immm	0.9882	target
RbS	mp-558071	exp.	Immm	0.0019	target
RbS	mp-1179684	theo.	C2/m	0.9975	target
Sr3P4	mp-14288	exp.	Fdd2	0.0000	target
Sr3P4	mp-682953	theo.	P1	0.0081	target
SrF2	mp-1102240	exp.	Pnma	0.0646	target
SrF2	mp-1102911	theo.	Pnma	0.0267	target
SrN	mp-1058586	exp.	Fm-3m	0.4343	target
SrO2	mp-726705	exp.	Pnma	0.0096	target
SrO2	mp-2763179	theo.	Pnma	0.0110	target
SrS	mp-10627	exp.	Pm-3m	0.2996	target

F extension4: Case-1 Decomposition Reactions

All 89 decomposition-into-elements reactions ($A_a B_b \rightarrow a A + b B$, standard-state elemental references), balanced by `reaction_icohp.py`, with ΔICOHP per atom and the endobondic/exobondic label computed by `reaction_analysis` (products – reactants convention — note this is the sign *opposite* `reaction_icohp.py`'s own `delta_icohp_per_atom` column reported elsewhere in this project; see `delta.py`'s docstring). Auto-generated from `analysis/reaction_icohp_case1.csv` and `analysis/reaction_analysis_case1_full.csv` by `report/gen_appendix_extension4.py`.

Table 5: Decomposition-into-elements (case 1) reactions computed for the 89 **extension4** compounds, with ΔICOHP per atom (products – reactants convention, **reaction_analysis**) and the corresponding endobondic/exobondic label.

Reaction (balanced)	$\Delta\text{ICOHP}/\text{atom}$ (eV)	Label
0.2 Ba ₅ P ₄ → Ba + 0.8 P	-0.9590	exobondic
BaCl ₂ → Ba + Cl ₂	-0.2244	exobondic
BaF ₃ → Ba + 1.5 F ₂	-0.8680	exobondic
BaF ₃ → Ba + 1.5 F ₂	-0.2493	exobondic
BaN ₂ → Ba + N ₂	-8.6248	exobondic
BaO → Ba + 0.5 O ₂	-0.3221	exobondic
BaO → Ba + 0.5 O ₂	-1.6129	exobondic
BaS ₃ → Ba + 3 S	-0.6233	exobondic
0.3333 Be ₃ N ₂ → Be + 0.3333 N ₂	-0.0746	exobondic
0.3333 Be ₃ N ₂ → Be + 0.3333 N ₂	0.3960	endobondic
0.3333 Be ₃ P ₂ → Be + 0.6667 P	-0.2111	exobondic
0.3333 Be ₃ P ₂ → Be + 0.6667 P	-0.1795	exobondic
BeCl ₂ → Be + Cl ₂	1.8204	endobondic
BeCl ₂ → Be + Cl ₂	-5.7807	exobondic
BeF ₂ → Be + F ₂	3.1514	endobondic
BeF ₂ → Be + F ₂	-0.6816	exobondic
BeO → Be + 0.5 O ₂	2.7864	endobondic
BeO → Be + 0.5 O ₂	4.4391	endobondic
0.3333 Ca ₃ N ₂ → Ca + 0.3333 N ₂	-4.4881	exobondic
0.3333 Ca ₃ N ₂ → Ca + 0.3333 N ₂	-4.4898	exobondic
CaCl ₂ → Ca + Cl ₂	-0.3199	exobondic
CaCl ₂ → Ca + Cl ₂	-0.2208	exobondic
CaF ₂ → Ca + F ₂	-0.2227	exobondic
CaF ₂ → Ca + F ₂	-0.1814	exobondic
CaO → Ca + 0.5 O ₂	-2.2127	exobondic
CaO → Ca + 0.5 O ₂	-2.9971	exobondic
0.5 Cs ₂ S → Cs + 0.5 S	0.9554	endobondic
CsCl → Cs + 0.5 Cl ₂	0.8830	endobondic
CsF → Cs + 0.5 F ₂	3.8464	endobondic
CsN ₃ → Cs + 1.5 N ₂	-2.2903	exobondic
CsN ₃ → Cs + 1.5 N ₂	-1.6664	exobondic
CsO ₂ → Cs + O ₂	0.0666	endobondic
CsO ₂ → Cs + O ₂	0.7743	endobondic
CsP ₇ → Cs + 7 P	-0.2978	exobondic
CsP ₇ → Cs + 7 P	-0.2667	exobondic
0.5 K ₂ O → K + 0.25 O ₂	-0.4357	exobondic
0.5 K ₂ O → K + 0.25 O ₂	-1.5638	exobondic
0.5 K ₂ S → K + 0.5 S	-0.3359	exobondic
KCl → K + 0.5 Cl ₂	0.7616	endobondic
KF ₃ → K + 1.5 F ₂	-1.1511	exobondic
KF ₃ → K + 1.5 F ₂	-1.1505	exobondic
KN ₃ → K + 1.5 N ₂	-5.9991	exobondic
KP → K + P	-2.1600	exobondic
KP → K + P	-2.1366	exobondic
0.5 Li ₂ O → Li + 0.25 O ₂	2.3366	endobondic
0.5 Li ₂ O → Li + 0.25 O ₂	1.9481	endobondic
0.5 Li ₂ S → Li + 0.5 S	1.8505	endobondic
0.5 Li ₂ S → Li + 0.5 S	1.5052	endobondic
0.3333 Li ₃ N → Li + 0.1667 N ₂	0.6165	endobondic
LiCl → Li + 0.5 Cl ₂	2.8030	endobondic
LiCl → Li + 0.5 Cl ₂	2.5005	endobondic

Reaction (balanced)	$\Delta\text{ICOHP}/\text{atom}$ (eV)	Label
$\text{LiF} \rightarrow \text{Li} + 0.5 \text{ F}_2$	4.4673	endobondic
$\text{LiF} \rightarrow \text{Li} + 0.5 \text{ F}_2$	4.5131	endobondic
$\text{LiP}_5 \rightarrow \text{Li} + 5 \text{ P}$	0.9043	endobondic
$\text{LiP}_5 \rightarrow \text{Li} + 5 \text{ P}$	0.8692	endobondic
$\text{MgCl}_2 \rightarrow \text{Mg} + \text{Cl}_2$	0.1568	endobondic
$\text{MgCl}_2 \rightarrow \text{Mg} + \text{Cl}_2$	-0.0862	exobondic
$\text{MgF}_2 \rightarrow \text{Mg} + \text{F}_2$	0.0015	endobondic
$\text{MgF}_2 \rightarrow \text{Mg} + \text{F}_2$	0.0229	endobondic
$\text{MgO} \rightarrow \text{Mg} + 0.5 \text{ O}_2$	-1.4179	exobondic
$\text{MgO} \rightarrow \text{Mg} + 0.5 \text{ O}_2$	-1.0763	exobondic
$\text{MgS} \rightarrow \text{Mg} + \text{S}$	-0.3401	exobondic
$\text{MgS} \rightarrow \text{Mg} + \text{S}$	-0.4163	exobondic
$0.5 \text{ Na}_2\text{Cl} \rightarrow \text{Na} + 0.25 \text{ Cl}_2$	-0.3965	exobondic
$0.5 \text{ Na}_2\text{O} \rightarrow \text{Na} + 0.25 \text{ O}_2$	-1.3057	exobondic
$0.5 \text{ Na}_2\text{O} \rightarrow \text{Na} + 0.25 \text{ O}_2$	-1.5619	exobondic
$\text{NaN}_3 \rightarrow \text{Na} + 1.5 \text{ N}_2$	-1.5201	exobondic
$\text{NaN}_3 \rightarrow \text{Na} + 1.5 \text{ N}_2$	-5.4248	exobondic
$\text{NaS} \rightarrow \text{Na} + \text{S}$	-0.4911	exobondic
$\text{NaS} \rightarrow \text{Na} + \text{S}$	-2.3999	exobondic
$0.5 \text{ Rb}_2\text{O} \rightarrow \text{Rb} + 0.25 \text{ O}_2$	0.7235	endobondic
$0.5 \text{ Rb}_2\text{O} \rightarrow \text{Rb} + 0.25 \text{ O}_2$	-0.7592	exobondic
$\text{RbCl} \rightarrow \text{Rb} + 0.5 \text{ Cl}_2$	1.7924	endobondic
$\text{RbF} \rightarrow \text{Rb} + 0.5 \text{ F}_2$	3.9244	endobondic
$\text{RbF} \rightarrow \text{Rb} + 0.5 \text{ F}_2$	1.9497	endobondic
$\text{RbN}_3 \rightarrow \text{Rb} + 1.5 \text{ N}_2$	-1.3478	exobondic
$\text{RbN}_3 \rightarrow \text{Rb} + 1.5 \text{ N}_2$	-3.5636	exobondic
$\text{RbP} \rightarrow \text{Rb} + \text{P}$	-1.9315	exobondic
$\text{RbP} \rightarrow \text{Rb} + \text{P}$	-1.9104	exobondic
$\text{RbS} \rightarrow \text{Rb} + \text{S}$	-0.3217	exobondic
$\text{RbS} \rightarrow \text{Rb} + \text{S}$	-2.7347	exobondic
$0.3333 \text{ Sr}_3\text{P}_4 \rightarrow \text{Sr} + 1.333 \text{ P}$	-1.0472	exobondic
$0.3333 \text{ Sr}_3\text{P}_4 \rightarrow \text{Sr} + 1.333 \text{ P}$	-1.0793	exobondic
$\text{SrF}_2 \rightarrow \text{Sr} + \text{F}_2$	-0.5421	exobondic
$\text{SrF}_2 \rightarrow \text{Sr} + \text{F}_2$	-0.1481	exobondic
$\text{SrN} \rightarrow \text{Sr} + 0.5 \text{ N}_2$	-4.5395	exobondic
$\text{SrO}_2 \rightarrow \text{Sr} + \text{O}_2$	-1.9205	exobondic
$\text{SrO}_2 \rightarrow \text{Sr} + \text{O}_2$	-1.7522	exobondic
$\text{SrS} \rightarrow \text{Sr} + \text{S}$	-0.1088	exobondic